Development of Porous Ceramics Using Orange Peel Waste for Thermal Insulation Applications

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Abstract

This research investigates the development of porous ceramic materials by incorporating orange peel waste (OPW) as a natural pore-forming agent. Porous ceramics have wide use in thermal insulation, sound absorption, and lightweight structural applications. In this study, a batch composition consisting of Ball clay, China clay, feldspar, quartz, talc, and CMC binder was used. Orange peel waste (5 wt%) was added to introduce controlled porosity during firing.

The samples were fabricated using a conventional ceramic processing route and sintered at 900–1100 °C. The physical and mechanical properties were evaluated through Archimedes density, water absorption, porosity, impact strength, bending strength, and thermal conductivity (Lee's Disc method). Results showed that increasing OPW content significantly increased porosity and reduced density and thermal conductivity. This study demonstrates the feasibility of converting agricultural waste into valueable porous ceramic materials for sustainable thermal insulation applications.

Keywords: Porous ceramics, Orange peel waste, Pore-forming agent, Sintering, Thermal properties.

Introduction

Porous ceramics are widely used in thermal insulation, heat shields, filtration, and catalyst supports due to their lightweight structure and low thermal conductivity. Traditionally, controlled porosity is achieved using organic pore formers such as sawdust, rice husk, and polymer beads.

Orange peel waste (OPW) is an abundant agricultural waste rich in cellulose, hemicellulose, and essential oils. Upon heating, it decomposes, releasing volatiles that generate well

distributed pores. Converting OPW into pore formers provides a sustainable and economical route for developing green ceramic materials.

The objective of this research is:

- To develop porous ceramics using OPW as a pore-forming agent
- To evaluate the effect of OPW on physical, mechanical, and thermal properties
- To identify the optimum composition for thermal insulation

Materials and Batch Composition

The base batch calculation (for 120 g) is:

Raw materials	Percentage (%)	Weight(gm)
feldspar	42	50.4
China clay	15	18
Ball clay	28	33
Quartz	7	8.4
Talc	3	3.6
Waste	5	6
CMC	3-5	

Experimental Procedure

The ceramic mixture used in this research followed the batch calculation taken from the provided formulation. For every 120-gram batch, the raw materials consisted of 50.4 g of feldspar, 33.6 g of ball clay, 18 g of china clay, 8.4 g of quartz, and 3.6 g of talc. A constant waste allowance of 5% (6 g) was included and was not varied at any stage of the experiment. To ensure proper plasticity and binding during shaping, a CMC binder (approximately 4 g) was added at a fixed percentage.

The orange peel waste was first washed, dried at 80°C for 24 hours, ground into fine powder, and screened through a 150-micron sieve. This powder was blended thoroughly with the ceramic raw materials. CMC mixed in water was introduced to form a uniform and workable mass, which was then granulated and compacted into both cylindrical and rectangular test specimens using a hydraulic press at 150 MPa.

The shaped samples were dried at 110°C for 12 hours to eliminate moisture before firing. A controlled heating rate of 5°C per minute was applied, and sintering was carried out at 900°C, 1000°C, and 1100°C. As the temperature increased, the organic matter from the orange peel burned away, generating pores throughout the ceramic structure.

After sintering, the samples underwent a full set of physical, mechanical, and thermal tests, including Archimedes density measurement, water absorption, apparent porosity, bending strength, impact strength, and thermal conductivity using Lee's disc. All parameters were calculated using standard formulas to determine how the varying OPW additions influenced the performance of the final ceramic bodies.

Fig: Orange Peels

Fig: Prepared sample

Impact strength test

The impact test determines the amount of energy a ceramic specimen can absorb when subjected to sudden shock loading. Ceramics are brittle materials, fail rapidly under impact loading. Therefore, the material's ability to withstand such impact forces is an important indicator of toughness. In this investigation, a Charpy impact machine was used to measure the absorbed energy when the pendulum striker fractured the specimen. The energy recorded by the machine is then used to compute the impact strength. For this study, the impact strength was calculated using the relationship that correlates the absorbed energy with the square of the specimen thickness. The formula applied is:

$$Impact\ Strength = \frac{12I}{t^2}$$

where I represent the absorbed impact energy in joules (J), and t is the thickness of the specimen at the point of impact. This formula reflects how the fracture resistance of brittle materials is strongly influenced by specimen thickness.

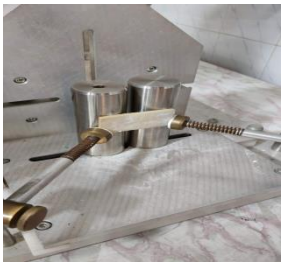


Fig: before testing **fig: set up of testing** **fig: after testing**

Sample Calculation

The thickness of the ceramic specimen is 0.219 inch and the Charpy machine records a load 0.03, then the impact strength is calculated as:

$$\textit{Impact Strength}=\frac{12I}{t^2}=\frac{12\times0.03}{(0.219)^2}=7.506\ psi$$

This value indicates the energy absorption capability of the material before fracture. Samples containing higher amounts of orange peel waste typically show lower values of Thermal conductivity due to the increased porosity that weakens the ceramic matrix.

Bending (Flexural) Test

The bending test, conducted using a three-point flexural setup, determines how much stress a ceramic specimen can withstand before breaking under static loading. This test is particularly valuable because bending conditions are common in practical applications, and ceramics typically fail without plastic deformation. During the test, the specimen is supported at two ends while a load is applied at the centre. The bending stress at fracture is referred to as the flexural strength. The test assumes the specimen behaves elastically until failure. The flexural strength for a rectangular specimen is calculated using the standard formula:

$$\sigma=\frac{3FL}{2bd^2}=\frac{3\times156.8\times40}{2\times32.44\times(5.40)^2}=9.945\ mpa$$

where
F = fracture load (N)= 16 kg =156.8 N
L = span length = 40 mm
b = sample width = 32.44 mm
d = sample thickness = 5.40 mm

Porous samples containing more OPW typically produce lower bending strength due to the increased number of voids that initiate crack propagation and failure of the body.

Fig: before testing

Fig: Testing setup

Fig: After testing

Water Absorption Test

The water absorption test measures the amount to which fired ceramic samples can absorb water or moisture, which is directly related to the quantity and connectivity of pores in the material. The test involves weighing the dry sample, then immersing it in water for 24 hours, and finally weighing it again in its saturated state. The increase in mass indicates the amount of water absorbed through open pores. If porosity in the sample increases, the percentage of water absorption also increases. Water absorption is expressed as a percentage of the dry mass and is calculated by:

$$\textit{Water Absorption (\%)}=\frac{W_{wet}-W_{dry}}{W_{dry}}\times 100=\frac{14.11-14.10}{14.10}\times 100=0.07\%$$

Fig: soak into water for 24 hours

Density Test

Density of the ceramic sample was determined using Archimedes water immersion method by measuring the increase in weight after saturation. Firstly, we measured the dry weight of the sample, then allowed it suspended into water and took the measure of weight in this condition. Finally, we took the saturated weight after soaking. This test indicates the open porosity and moisture uptake behaviour of the ceramic body.

Given:

Dry weight of sample in air, Wd = 3.011 g

Suspended weight of sample in water, Ws = 1.69 g

Saturated weight of sample in air, W_{sat} = 3.044 g

Density of water, ρ_w = 1.0 $g/(cm^3)$

Formula:

Bulk Density = $\frac{Wsat - Ws}{Wd} \times \rho_w$

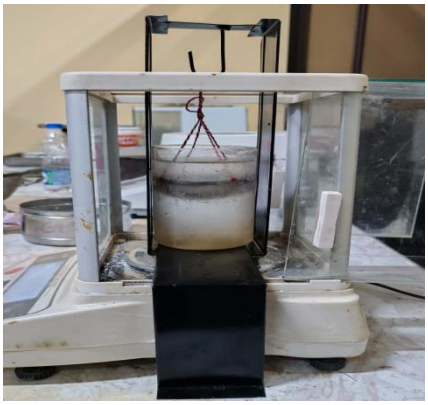
Calculation:

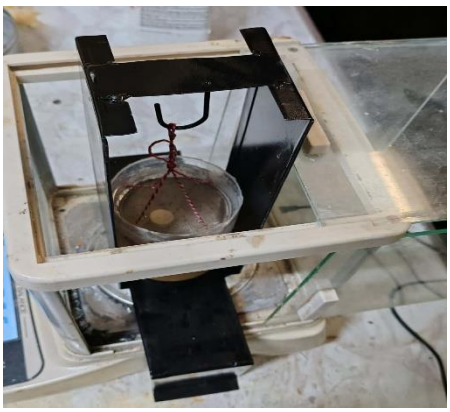
W_{sat} - W_s = 3.044 - 1.69 = 1.354 g

Bulk Density = $\frac{3.011}{1.354} \times 1$ = 2.22 $g/(cm^3)$

Result:

Bulk Density of the ceramic sample = 2.22 g/cm³





Bulk density measurement using the Archimedes method.

Thermal conductivity

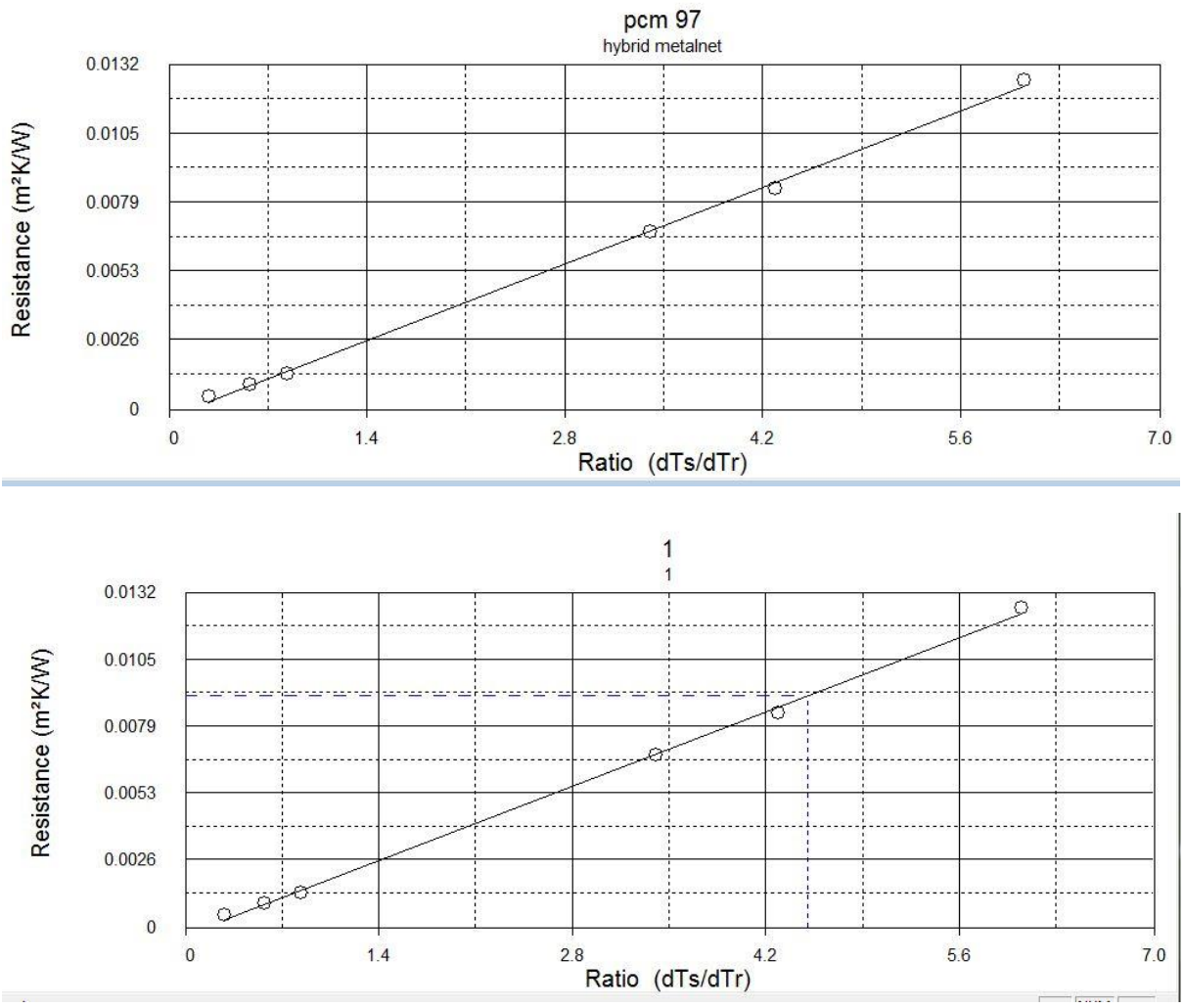
Thermal conductivity is the measure of how much heat flows through a material per unit time per unit area at a definite temperature gradient. A material having an increasing amount of porosity, traps more air in those pores and providing thermal insulation, lowers thermal conductivity.

Method

The thermal conductivity test was carried out using a radial heat transfer technique. A thermal resistance versus temperature ratio curve was obtained under steady-state conditions to determine the thermal conductivity of the ceramic sample.

Experimental Data

During the test, the temperature ratio and corresponding thermal resistance values were recorded after achieving steady-state heat flow.



Thermal resistance–temperature ratio curves from radial thermal conductivity testing.

Calculation of Thermal Conductivity

The thermal conductivity of the sample was calculated from the obtained thermal resistance using the relation:

Thermal Conductivity, k = $\frac{L}{R}$

where
L = thickness of the sample (m)
R = thermal resistance obtained from the curve (m^2K/W)

Collected data from the curve

Point 1:
$x_1 = 1.4$
$R_1 = 0.0025\ m^2K/W$

Point 2:
$x_2 = 2.1$
$R_2 = 0.00397\ m^2K/W$

Thickness of sample:
L = 4.32 mm = 0.00432 m

Thermal resistance is obtained from the slope of the straight line:

$R = \frac{R2-R1}{X2-X1}$

Substituting values, $R = \frac{0.00397-0.0025}{2.1-1.4}$ = 0.00210m2K/W

Calculate thermal conductivity

Formula:

Thermal Conductivity $k = \frac{L}{R} = \frac{0.00432}{0.00210}$ = 2.06 W/m·K

Result:

The thermal resistance was obtained from the slope of the linear thermal resistance versus temperature ratio curve. Using the radial heat transfer relation, the thermal conductivity of the ceramic sample was calculated as **2.06 W/m·K**.

Porosity Test

Firstly, dry weight of the sample was taken. Then ceramic samples were boiled in water more or less 1 hour. The water fills the open pores and the saturated weight of the sample was measured. The difference between saturated weight and dry weight will provide the weight of pore that holds water and the following shows the measurement of porosity:-

Given:

Dry weight of the sample, $W_d = 7.996$ g

Saturated weight of the sample, W_{sat} = 8.93 g

Formula:

$$\textbf{Porosity} = \frac{(Wsat - Wd)}{Wd} \times 100\%$$

Calculation:

$$\text{Porosity} = \frac{(8.93 - 7.996)}{7.996} \times 100\ \%$$

$$= 11.68\ \%$$

Result:

Porosity of the ceramic sample = 11.68 %

Fig: Porosity test

Discussion

From the experimental study, it is revealed that the orange peel waste can be effectively utilized as a natural pore former for the ceramic samples, thereby significantly impacting the physical, mechanical, and thermal properties. The decomposition of the orange peel waste is highly significant for creating porosity.

The recorded decrease in the bulk density value for both types of fired samples is mainly due to the burning of organic materials present in orange peel waste. The burning of OPW material results in the production of volatile gases, causing porosity in the desired ceramics. These pores are not fully closed through partial vitrification by the applied conditions for sintering, thus validating the creation of lightweight ceramics, which is useful for making insulator materials.

The results of water absorption and apparent porosity support these characteristics. The porosity result of 11.68% shows the existence of open pores created from the OPW burnout process. The moderate water absorption also shows a balance in the pore structure. The existence of a balanced pore structure is ideal for materials in the form of thermal insulation. The porosity here helps in the retention of air to create a thermal insulation material.

The mechanical properties, bending, and impact strength showed a declining trend with increasing porosity. This is expected, as the pores in the materials are locations for the initiation of cracks due to the stress concentration in them. Although the porous ceramics are not useful for structural purposes, the retention in mechanical strength is adequate for thermal insulation.

An important result of the present work is the enhancement in the thermal insulation due to the rise in porosity. The obtained thermal conductivity of 2.06 W/m·K verifies that the air-filled pores in the ceramic matrix resist heat flow to a certain extent. The decreased thermal conductivity also demonstrates that the generated porous ceramics from OPW can be applicable as insulating bricks, kiln lining, or lightweight thermal insulation.

In general, the experimental results reveal that the orange peel waste addition leads to a desirable combination of pore characteristics, mechanical strength attributes, and thermal insulation properties. The data supports the validity of using agricultural refuse to create high-quality porous ceramic materials.

Conclusion:

In this experiment, porous ceramics were successfully prepared using orange peel waste, which acted as a natural pore former, and the results showed an increase in its water absorption and decrease in its bulk density and thermal conductivity, making it ideal for thermal insulation. However, the bending strength and impact resistance of the ceramics decreased with an increase in porosity, and the results, although lower, are acceptable, making it ideal for non-structural thermal insulation applications. Furthermore, this experiment proved that orange peel waste can be used for making light, cheap, and eco-friendly porous ceramics for thermal insulation. Additionally, the experiment proved that porous ceramics have applications in construction, which can potentially help in reducing energy consumption and, in turn, the levels of carbon emissions, which pose one of the most significant threats to the environment..

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